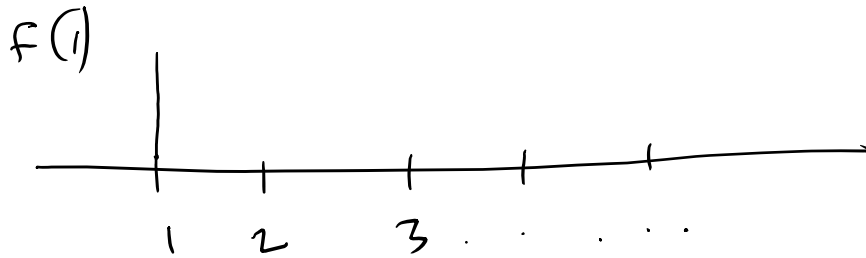


$$\sum_{x=1}^{\infty} x \boxed{f(x)} = E(X)$$

→ probabilities

= weighted sum of probabilities



$$f(1) = P(X=1)$$

$$f(2) = P(X=2)$$

$$\sum x f(x) = \sum_{i=1}^{\infty} i \underline{P(X=i)}$$

Example - Suppose we toss a die.

Let X = the number appearing on the top face of the die

$$E(X) = \sum x_i f(x_i)$$

Assuming the die is a uniform die

$$f(x_i) = \frac{1}{6} \text{ for all } i$$

$$E(X) = 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + 3\left(\frac{1}{6}\right) + \dots + 6\left(\frac{1}{6}\right)$$

$$E(X) = 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + \dots + 6\left(\frac{1}{6}\right) \quad (6)$$

$$(1 + 2 + 3 + \dots + 6) \frac{1}{6} = \frac{6 \times 7}{2} \times \frac{1}{6}$$

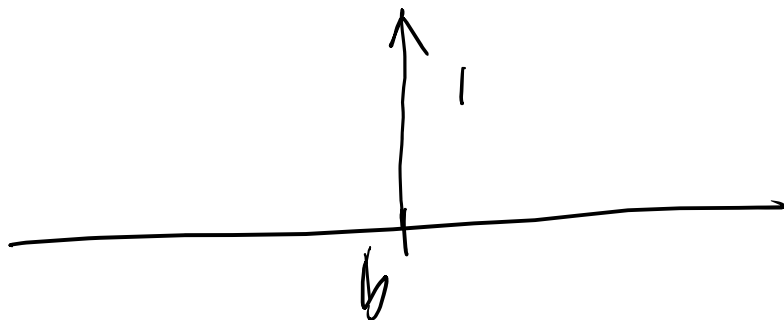
$$= \underline{3.5}$$

$$\begin{aligned}
 \text{Var}(X) &= E((X - \mu)^2) \\
 &= \sum (X - \mu)^2 f(X) \\
 &= \sum (X^2 - 2X\mu + \mu^2) f(X) \\
 &= \underbrace{\sum X^2 f(X)}_{\substack{\downarrow \\ E(X^2)}} - 2 \sum X \mu f(X) + \mu^2 \underbrace{\left(\sum f(X) \right)}_{=1} \\
 &= E(X^2) - 2\mu \underbrace{\sum X f(X)}_{\substack{\rightarrow \mu}} + \mu^2
 \end{aligned}$$

$$\begin{aligned}
 &= E(X^2) - (2\mu)(\mu) + \mu^2 \\
 &= E(X^2) - \mu^2
 \end{aligned}$$

$$\boxed{\text{Var}(X) = E(X^2) - E^2(X)}$$

$V(b) = 0$ means b is a constant



$\sum (X - \mu)^2 f(X) > 0$ when X takes different values.

$$V(aX) = a^2 V(X)$$

$$V(aX) = E((aX)^2) - E^2(aX)$$

$$= \sum (aX)^2 f(X)$$

$$a^2 \sum X^2 f(X) = a^2 E(X^2)$$

$$E(aX) = \sum aX f(X) = a \sum X f(X) = a E(X)$$

$$E^2(aX) = a^2 E^2(X)$$

$$V(aX) = E((aX)^2) - E^2(aX)$$

$$\begin{aligned}
 &= (1 - \rho^2) / 1 \\
 &= a^2 E(X^2) - a^2 E^2(X) \\
 &= a^2 (E(X^2) - E^2(X)) \\
 &= \underline{a^2 \text{Var}(X)}
 \end{aligned}$$

What about $V(aX + b)$?

$$\sum (aX + b)^2 f(x) = \underline{E((aX + b)^2)}$$

$$= \sum (a^2 X^2 + 2aXb + b^2) f(x)$$

$$= a^2 \sum X^2 f(x) + 2ab \sum X f(x) + b^2 \sum f(x)$$

$$= a^2 E(X^2) + 2ab E(X) + b^2$$

$$a^2 (E(X^2) - E^2(X))$$

$$+ a^2 E^2(X) + 2ab E(X) + b^2$$

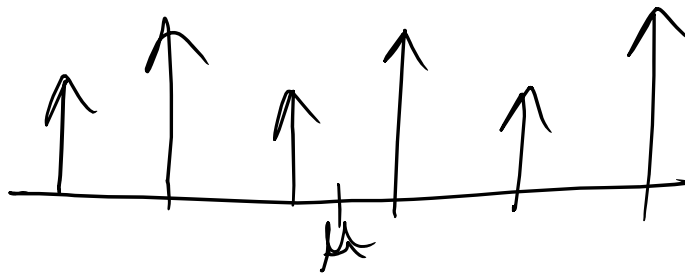
$$= a^2 \text{Var}(X) + (aE(X) + b)^2$$

$$\text{Var}(aX + b) = \underline{E((aX + b)^2)} - \underline{E^2(aX + b)}$$

$$a^2 \text{Var}(X) + (aE(X) + b)^2 - (aE(X) + b)^2$$

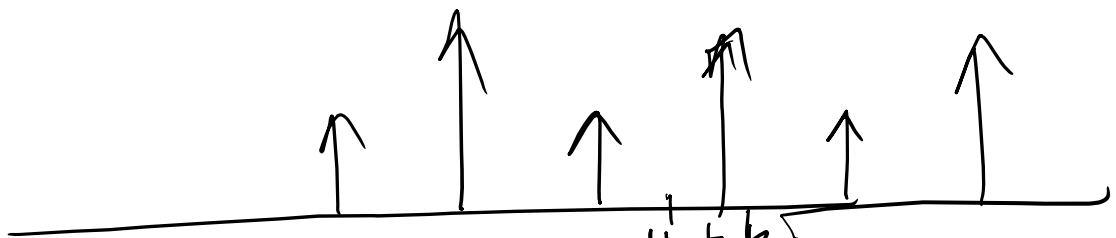
$$- (aE(X) + b)$$

$$\boxed{\text{Var}(aX + b) = a^2 \text{Var}(X)}$$



X $\text{Var}(X)$

$X + b$ is another random variable.



just shifted X

$\text{Var}(X + b) = \text{Var}(X)$

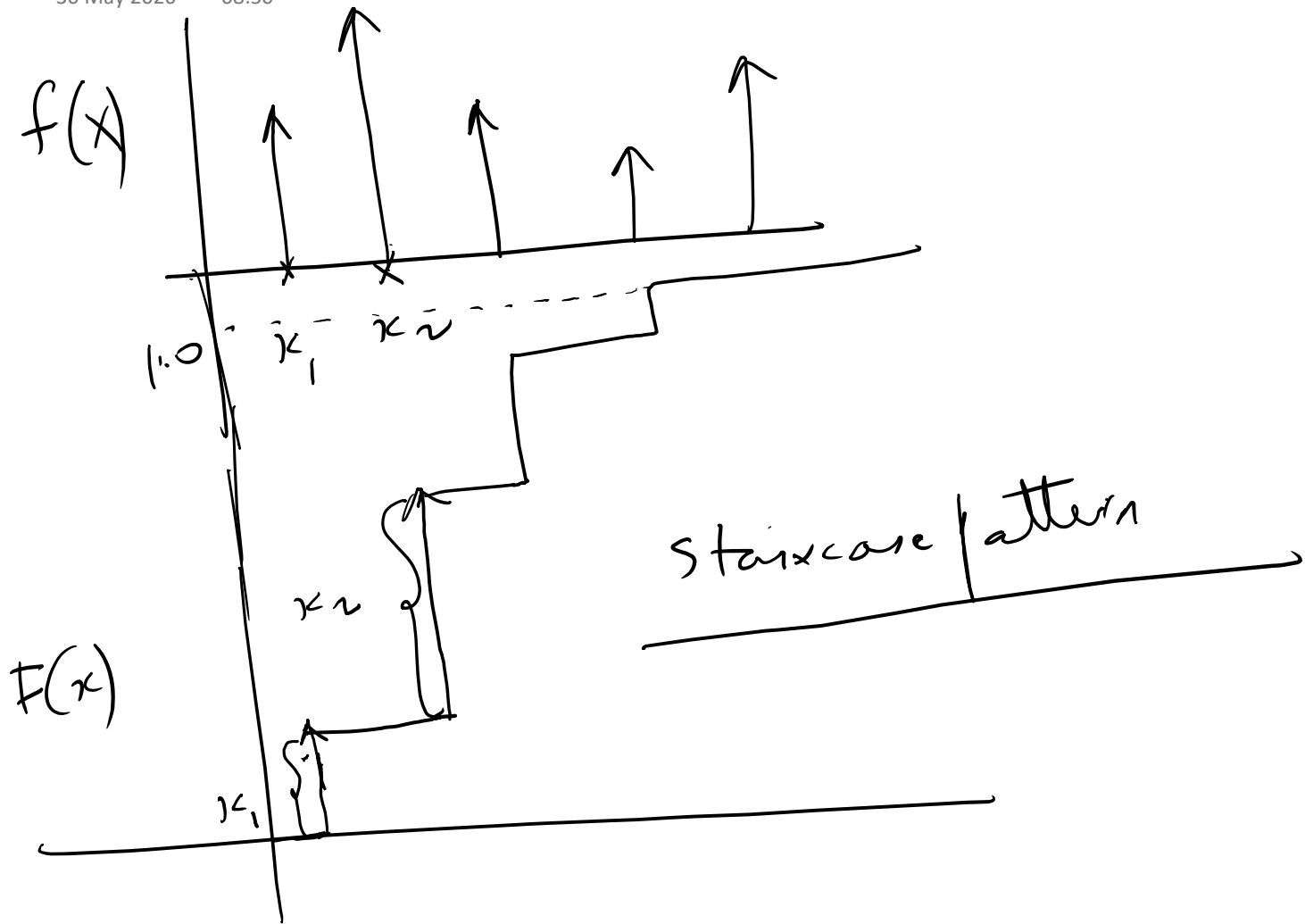
$$F(x_2) = P(X \leq x_2)$$

But $\{X \leq x_2\}$ denotes the

events $\{X = x_1\} \cup \{X = x_2\}$

mutually exclusive events

$$P(\{X = x_1\} \cup \{X = x_2\}) = P(\{X = x_1\}) + P(\{X = x_2\})$$



Die example

$$P\{X=2\} = \frac{1}{36} \rightarrow \text{why?}$$

When two die are tossed, what are the outcomes?

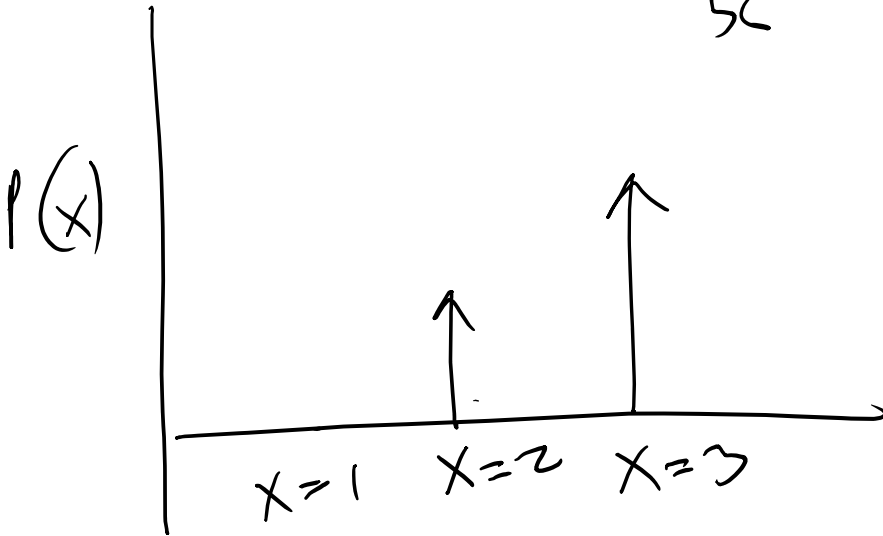
$$\left[\begin{array}{cccc} (1,1) & (1,2) & (1,3) & \dots & (1,6) \\ (2,1) & (2,2) & \dots & \dots & (2,6) \\ (3,1) & (3,2) & \dots & \dots & (3,6) \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ (6,1) & (6,2) & \dots & \dots & (6,6) \end{array} \right] = \underline{36.}$$

36 outcomes, all equally likely
 → each individual outcome
 has probability $\frac{1}{36}$

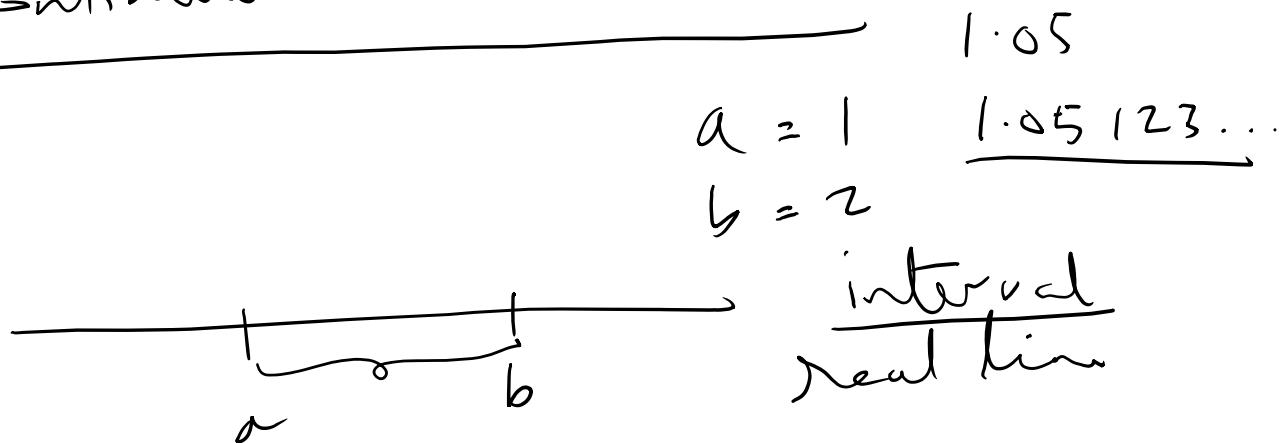
$$P\{X=3\} = (1,2), (2,1) \text{ are the possible outcomes}$$

$$\{X = 4\} = (1, 3), (3, 1), (2, 2)$$

$$\frac{1}{36} + \frac{1}{36} + \frac{1}{36} = \frac{3}{36}$$



Continuous Random Variable



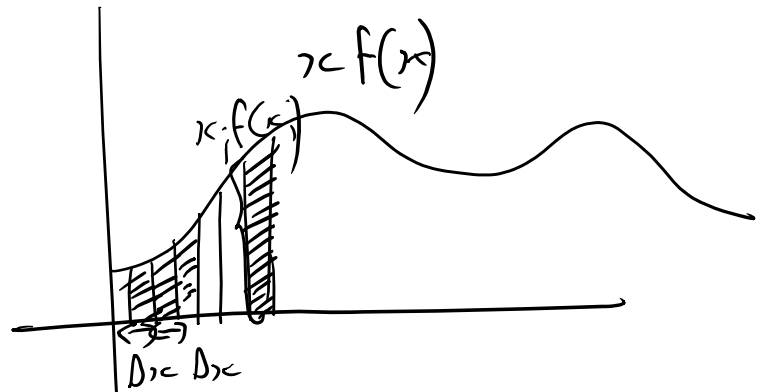
X can take any value between a & b .

→ there are infinite possibilities

We cannot talk about $P(X = x)$
for a continuous random
variable

$P(X = x)$ will be 0

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$



$$\sum x_i f(x_i) \Delta x$$

area of each strip = (height of strip)
(width of the strip)

$$\sigma^2 = V(X) = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx$$

$$\int_{-\infty}^{\infty} (x^2 - 2x\mu + \mu^2) f(x) dx$$

$$\int_{-\infty}^{\infty} x^2 f(x) dx$$

$$- 2\mu \int_{-\infty}^{\infty} x f(x) dx$$

like $\sum x_i^2 f(x_i) + \mu^2 \int_{-\infty}^{\infty} f(x) dx$ $\sum f(x_i)$

$$\begin{aligned} & E(x^2) - 2\mu E(x) + \mu^2 \\ &= E(x^2) - 2\mu^2 + \mu^2 \\ &= \underline{E(x^2) - \mu^2} \end{aligned}$$

$\sum_{i=1}^m \sum_{j=1}^n \rightarrow$ all possible combinations
 of i & j , i.e. all
 $m \cdot n$ possible combinations

$(i=1, j=1)$
 $(i=1, j=2)$
 $(i=1, j=3)$
 $(i=1, j=n)$

$(i=2, j=1)$
 $(i=2, j=2)$
 $\vdots$
 $(i=2, j=n)$

$$\sum_{i=1}^m \sum_{j=1, j \neq i}^n$$

$$i=1, j=2$$

$$i=1, j=3$$

⋮

$$i=1, j=n$$

$$i=2, j=1$$

$$i=2, j=3$$

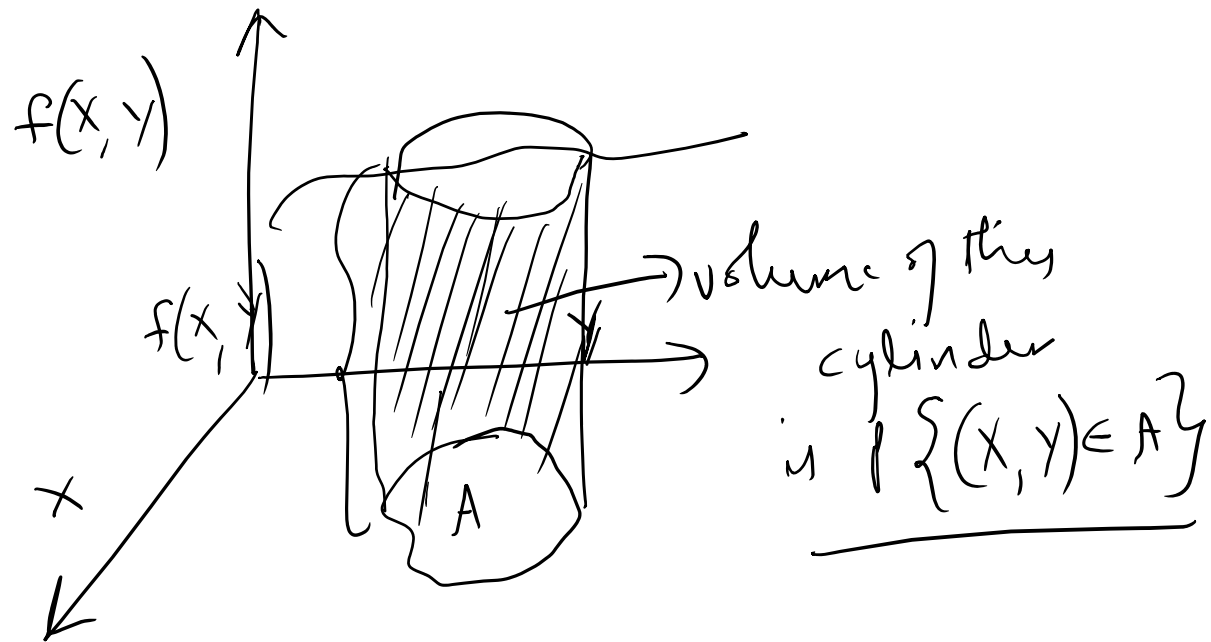
⋮

$$\sum_{i=1}^m \sum_{j>i}^n$$

$$\begin{array}{c} i=1, j=2 \\ \vdots \\ j=n \end{array} \bigg|$$

$$\begin{array}{c} i=2, j=3 \\ \vdots \\ j=n \end{array} \bigg|$$

$$\begin{array}{c} i=3, j=4 \\ \vdots \\ j=n \end{array}$$



$$\int \int_A f(x, y) \, dA = \int \int_A f(x, y)$$

if $f(x, y) = g(x)h(y)$, how do we know that $g(x)$ and $h(y)$ are marginal probability density functions?

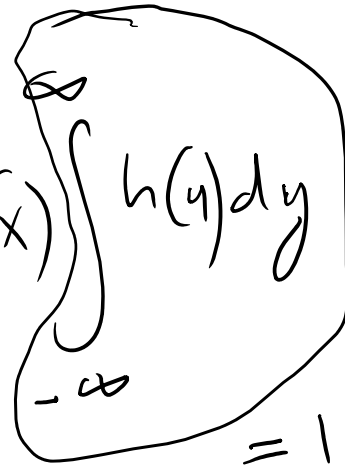
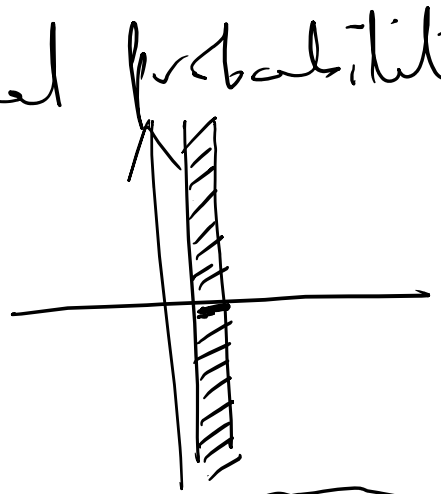
Let $f_{x\cdot}$ be the marginal probability density fn.

$$f_{x\cdot} = \int_{-\infty}^{\infty} f(x, y) dy$$

$$f_{x\cdot} = \int_{-\infty}^{\infty} g(x)h(y) dy = g(x) \int_{-\infty}^{\infty} h(y) dy$$

= 1

$$f_{x\cdot} = g(x)$$



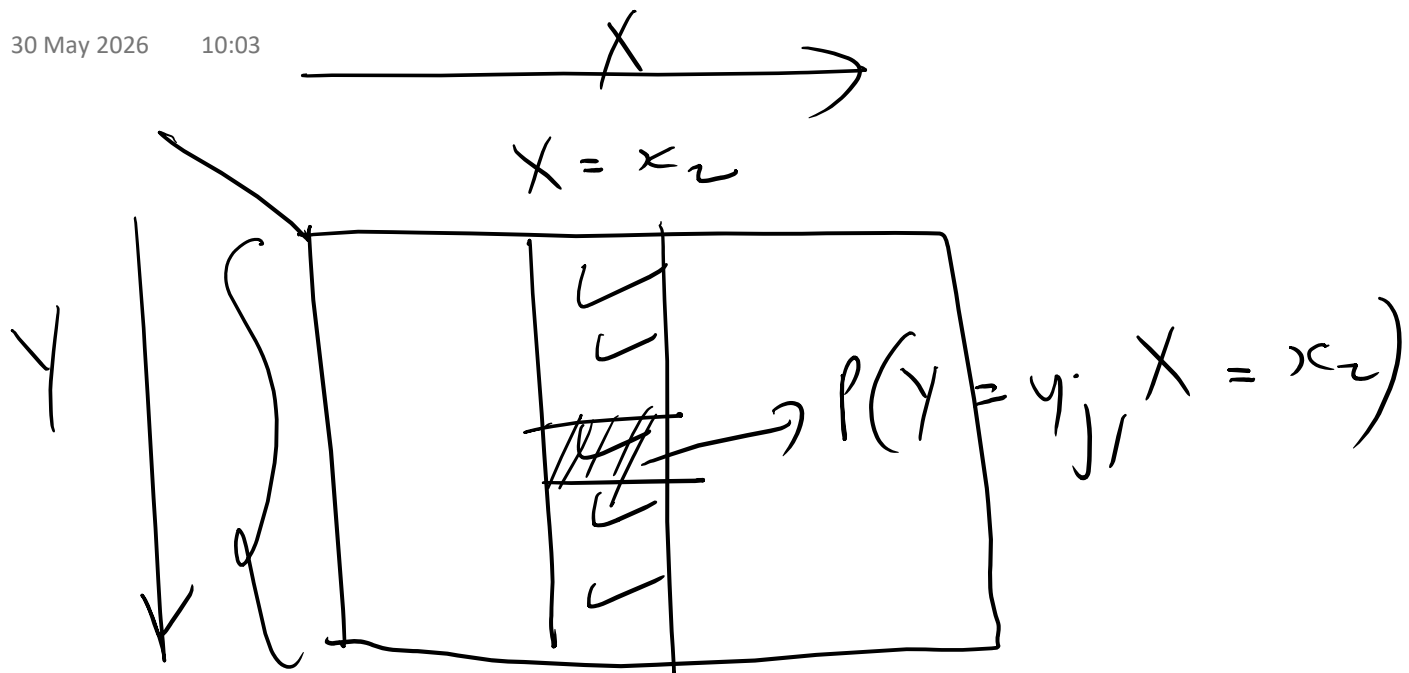
$X = x_1, X = x_2, \dots$

$Y = y_1$	P_{11}			
$Y = y_2$				
$\vdots$				

$$P(X = x_1 / Y = y_1) = \frac{P_{11}}{\sum_i P_{i1}} = \frac{P(X = x_1, Y = y_1)}{P_{.1}}$$

$$P(Y = y_1) = \sum_i P_{i1}$$

x can take any value, y is fixed



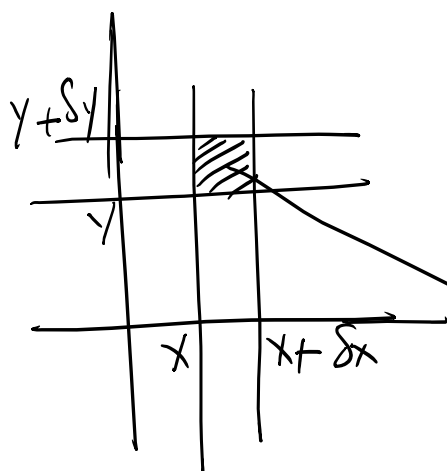
$$P(Y = y_j / X = x_2)$$

$$= \frac{P(Y = y_j, X = x_2)}{P(X = x_2)}$$

$$f(y/x) \delta y$$

$$P\{x \leq x \leq x + \delta x, y \leq y \leq y + \delta y\}$$

$$P\{x \leq x \leq x + \delta x\}$$



$$\frac{f_{xy}(x, y) \delta x \delta y}{f_x(x) \delta x}$$

$$\rightarrow P\{x \leq x \leq x + \delta x, y \leq y \leq y + \delta y\}$$

$$f(y/x) \delta y = \frac{f_{xy}(x, y) \delta x \delta y}{f_x(x) \delta x}$$

$$f(y/x) = \frac{f_{xy}(x, y)}{f_x(x)}$$